

Regulating Sustainability: the Interplay of Laws, Policies, Norms, and Design in Sustainable CSCW

Chiara Rossitto
Department of Computer and
Systems Sciences
Stockholm University
Stockholm, Sweden
chiara@dsv.su.se

Anton Poikolainen Rosén
Department of Computer and
Systems Sciences
Stockholm University
Stockholm, Sweden
anton.poikolainenrosen@dsv.su.se

Rob Comber
Department of Media Technology and
Interaction Design
KTH Royal Institute of Technology
Stockholm, Sweden
rcomber@kth.se

Stanley Joel Greenstein
Department of Law
Stockholm University
Stockholm, Sweden
stanley.greenstein@juridicum.su.se

Fatemeh Bakhshoudeh
Department of Media Technology and
Interaction Design
KTH Royal Institute of Technology
Stockholm, Sweden
fatemehb@kth.se

Lachlan Urquhart
School of Law; Institute for Design
Informatics
University of Edinburgh
Edinburgh, United Kingdom
lachlan.urquhart@ed.ac.uk

Susan Lechelt
Institute for Design Informatics
University of Edinburgh
Edinburgh, United Kingdom
susan.lechelt@ed.ac.uk

Abstract

This one-day workshop explores how encounters with regulatory aspects, before, during and after design, can shape the production of more sustainable futures. Within CSCW and Sustainable HCI, there have been repeated calls to develop more systemic approaches to design that include the political, institutional, and infrastructural aspects underpinning technology-mediated interventions towards sustainability. This workshop targets these calls by drawing attention to the methodological, analytical, and design challenges of considering regulatory and policy-making aspects that can shape both the design and the processes of designing digital technologies. It will be an opportunity to bring together CSCW, Sustainability, Law and other interested scholars, representatives of public institutions, environmental collectives, and diverse actors interested in investigating the "knots" of relations between technology designs, regulatory aspects, and sustainable practices. We plan to accept up to twenty contributions and run the workshop on-site.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CSCW Companion '25, Bergen, Norway
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1480-1/25/10
<https://doi.org/10.1145/3715070.3748290>

Keywords

Regulation, Sustainability, Law, Norms, Policy Knot, Design Process

ACM Reference Format:

Chiara Rossitto, Anton Poikolainen Rosén, Rob Comber, Stanley Joel Greenstein, Fatemeh Bakhshoudeh, Lachlan Urquhart, and Susan Lechelt. 2025. Regulating Sustainability: the Interplay of Laws, Policies, Norms, and Design in Sustainable CSCW. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3715070.3748290>

1 Introduction

Issues of environmental protection are intrinsically related to regulatory aspects. For example, the EU EcoDesign Regulations require both resilient hardware and long-term access to spare components [1]. Simultaneously, local laws can define pollution thresholds [24], mandate compliance with the right to repair digital devices [38], establish how digital platforms, meant to foster environmental care, are supposed to be used within urban waste infrastructure [8], or how environmental and climate data can inform changes in policy-making [9, 13, 31, 32]. However, despite repeated calls to encompass the political, institutional, [6, 8, 11], infrastructural [19, 22, 33], and relational [26, 31] aspects that more holistically scope projects oriented towards sustainability, the role of laws and other regulations is still rarely addressed within Sustainable HCI [15], and CSCW research more broadly. As argued by Urquhart et al. [38], understanding and addressing the relations between such systemic aspects and design requires an interdisciplinary perspective combining design, law, social science, and sustainability research.

Beyond sustainability [37, 38], HCI and CSCW studies on the platform economy [10, 18, 21], privacy and social media moderation [14] have emphasised concerns to better engage with the regulatory and policy aspects of design. However, the field still lacks conceptual frameworks, empirical understandings and design methodologies that investigate how regulations might affect design and, vice versa, how technological advances might push for regulatory changes. We see new ways of approaching technology design as essential for future work within SHCI, as both researchers and design practitioners (see [25]) increasingly face ethical aspects and concerns for environmental and social sustainability before, during, and after designing digital technologies. Jackson and colleagues [20] suggested the metaphor of the policy knot to illustrate how policy, design, and practices can be entangled and frame both technology design and use. Yet, a decade after being proposed, this aspect of design is still underexplored, while recent environmental challenges and technological advances call for a reinvigorated interest in the topic (e.g., AI-based applications¹ and the large energy and data infrastructures they rely on, cf [3, 7]).

This workshop provides opportunities for the development of future research directions for sustainability in CSCW research. It will enable focused and in-depth discussions on the concepts, empirical cases, and technological explorations that help articulate what *regulating sustainable design* means and might imply. The workshop will be interdisciplinary and open to researchers with various backgrounds (e.g., SCHI, Law and IT), practitioners, and other professionals interested in exploring the encounters between environmental sustainability, regulatory aspects, design, and designing. It targets the many actors (e.g., designers, industry representatives, think tanks, public institutions) concerned with environmental, economic, and social sustainability, and who are interested in developing participatory approaches and methods to explore tensions and possibilities in the innovation-regulation nexus [27]. The workshop's specific **goals** are to:

- Map conceptual, empirical, and technological examples of encounters with the law and other regulations before, during, and after design.
- Develop a nuanced vocabulary, metaphors, or conceptual frameworks to talk about different forms of encounters between regulatory aspects, design, and digitally mediated practices of sustainability.
- Envision participatory methods, design materials, and strategies that can facilitate multi-party engagement with regulatory aspects throughout design processes.
- Outline the political dynamics, power relations, ethical issues, interests, or historical reasons that can shape: *i*) how concerns for specific digital technologies gain legal attention, are framed, and contested; *ii*) the considerations and inclusion of regulatory aspects in design processes.
- Identify individual, collective, and organisational challenges that can hinder the reflection and inclusion of regulatory aspects in design processes.
- Outline ideas for future research agendas and collaborations.

¹For instance, concerning Generative AI, the EU AI Act (Regulation (EU) 2024/1689) mandates transparency for energy consumption during use, alongside accountability for computational resources in training and development (<https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>)

2 Background

Ten years ago, Jackson, Gillepsie, and Payette [20] argued for the need to better understand the mutual relations between policy, technology design, and technology-mediated practices, which they characterised through the metaphor of the 'policy knot'. The metaphor highlights the complex and shifting nature of such relations, and the need to include reflections about regulatory aspects early on in design processes, not only *after* digital technologies have been designed and deployed. Despite a renewed interest in regulation and its consequences over the last years [5, 8, 29, 37, 38], this aspect of design and designing still remains underexplored within HCI and CSCW research. Moreover, so are the frameworks, conceptualisations, and methodological approaches to explore the entanglements between digitally mediated practices, regulatory aspects, and how multi-party interests (from issues of governance to individuals' everyday practices) are configured, can be contested, and evolve over time. Roscam and Light [30] have illustrated how complying with specific legal requirements can be at odds with exploratory design processes, by being counterproductive to their time frame and the very participatory concerns of technology design. Other research [2] has instead illustrated the use of speculative scenarios to highlight how laws and policies can shape the design of AI-based systems for public spaces, while providing opportunities to contest and improve the design of AI systems. Directly related to the focus of this workshop, our work [37] has proposed the use of law-based design card as a way to elicit sustainability-centred requirements in the design of IoT devices.

Sustainability in HCI has been developed between what can be seen as different positions of regulation: 1) the normative values (e.g., care, repair, longevity) of design from within a sustainability perspective, arguably originating within HCI in Eli Blevis' [4] consolidation of design thinking in 'Sustainable Interaction Design', 2) the nexus of values implied in the cognitive model of the 'Resource Man', – e.g., control, measurably, rationality and choice [34] – and how to design for self-regulation of consumption, and 3) in the hyper-decontextualised global goals of sustainable development and decarbonisation (e.g., Paris Agreement) [17]. Dourish [11] has more specifically argued for a need to understand the role of national and international regulations in shaping transitions towards more sustainable futures, while warning that failing to recognise questions of political ecology, environmental justice, and governance limits HCI engagements with environmental sustainability. Relatedly, studies focused on the use of digital technologies (e.g., sensors, platforms) to collect environmental data and advocate for changes in governance [9, 31, 33] provide examples of how relations between Sustainable HCI projects and regulatory aspects can be configured to push for changes in policy and local governance.

In this workshop we seek to explore and expand these domains of regulations in relation to CSCW. For instance there has been fewer scholarship engaging with regulation that originates in markets (see [35]) or with the relations between markets and laws (e.g., concerns to stop greenwashing and misleading customers around eco-credentials of products (see the 2024/825/EC REF <https://eur-lex.europa.eu/eli/dir/2024/825/oj/eng>))

2.1 Topics of interest

Workshop participants can address a range of themes, including but not limited to the following:

Tensions between regulation and innovation. The relations to laws and other forms of regulations are not a new concern for the design of digital technologies, with regulatory fields such as legal informatics, consumer protection, employment and labour law, disability laws, or copyright protection being illustrative examples of the “regulation-innovation nexus” [27] – that is, the interplay between drivers of digital innovations and concerns for the framing of technology design within defined regulatory and legal frameworks. On the one hand, as illustrated in the case of a digital platform promoting environmental care and protection in waste disposal through on demand-labour [8], certain technologies did not exist when the law or other forms of regulation were made, and it is here that the law can become an obstacle to technological explorations, and the wide adoption of new technologies [16]. On the other hand, the articulation of concepts that are vague and imprecise, is a way for the law to counter the risk of becoming obsolete over a short period of time. This, in turn, can lead to potential uncertainty and an increased unwillingness to innovate, with concerns about contravening the law outweighing creative design processes, and explorations about the utility and enhanced values of novel digital technologies. We invite submissions that address the tension between regulation as potentially hindering innovation, on the one hand, and as protecting from adversarial effects, on the other. Focus can be placed, for example, on how this tension can be differently configured throughout technology design and its situated use. We additionally encourage submissions that discuss potential conflicts between essential socio-cultural values (e.g., environmental care, sustainability and social well-being, or other values we want to protect) and different drivers of technology design (e.g., growth, self-enhancement, profit), and between the framing of the law as a means for solving problems (when they arise) or as an hindrance to digital innovation.

Conceptual framings. *“Regulability is the capacity of governments to regulate behaviours within its proper reach”* [23, p.24]. While this definition is often –and simplistically – associated with law books and legislation (i.e., hard law), the concept of regulation is more nuanced. We suggest that unpacking its multi-faceted meanings can help CSCW to: i) better engage with regulatory aspects, and ii) reflect on the specific HCI and CSCW concerns (i.e., the role of code and design) that can already be seen as ways to regulate (constraint and enable) collective and individual behaviours. This entails highlighting the distinction between legislation, on the one hand, and regulation on the other, the former referring to hard law, with the latter referring to alternative regulatory tools, such as codes of conduct, standards or even technology itself. Legal scholar Lessig [23] has suggested four different modalities that can regulate peoples’ behaviours: social norms, architecture, the market, and law. Echoing a long-standing tradition within HCI, *social norms* refers to the role of socio-cultural aspects that shape what people do regardless of sanctions. The very formulations of sustainability in and through design can be connected to this point. *Architecture* relates to the code that defines the structure of digital environments

and what is allowed to do within –or with– them. The modality of *market* emphasises how market design and other economic factors can regulate a certain situation; business models promoting the short-term use and obsolescence by design of digital products are relevant examples of this modality. We invite submissions that bring attention to the conceptual nuances of what regulation means and entails, and how such understandings (e.g., concepts, metaphors) can inform SHCI and CSCW as research endeavours.

Relations to policy-making. Research has explored the role of digital technology and data to advocate for policy-making concerned with environmental issues – from maintaining biodiversity [36] to reducing pollution and environmental harm [9]. Our previous work [31, 32] has shown, for instance, how the design of digital platforms can afford the collection of litter data, enable the revisions of local sustainability policies, and help municipalities enforce regulation to limit urban pollution. Yet, it is unclear how policy-making processes are configured on the ground, and what collaborations and partnerships underlie them. Moreover, it is still unexplored what role interested actors and professional expertise (see [28]) can play in linking data to policy-making concerned with more sustainable futures. We invite empirical accounts and technological explorations that explore the interplay between technology-mediated acts of environmental care how they can fold into processes of policy-making.

Methodological explorations. We invite explorations of new methods and design tools to reflect on diverse forms of regulations before, during, and after design. While approaches, such as, ethnographic explorations or content analysis can help trace possible configurations of policy knots, we call for the development of novel participatory methods that can support multi-party engagement in reflecting, noticing, and concretely configuring relations between regulations, practices of sustainability, and design. There are also opportunities for more speculative methods, such as, design cards [37], design fictions [39], futuring workshops [12], and critical designs to support conversations among the many involved stakeholders, and about the diverse political, economic, or environmental protection interests that might shape their view and experience of regulations. We see the need here for methods to balance reflections on different regulation modalities and how they become entangled with the interests of different actors. We invite submissions that: i) illustrate how to concretely brainstorm, map, examine and compare various encounters with regulations when designing for sustainability; ii) support reflections on the different involved stakeholders, legal values and ethical tensions, issues of innovation and scale, and on the relations between digital explorations and regulatory aspects (e.g., the identification of regulatory gaps or laws to comply with).

3 Pre-workshop Plan

Authors are invited to submit a 2-4 page position paper following the ACM single-column format – either by using the Word or LaTeX template. To promote contributions that are relevant for practitioners (e.g., lawyers, activists, representatives of the public sector, designers, and other), we will also encourage the submission of alternative research and design materials, such as, design portfolios, short manifestos, pictorials, speculative designs, or design

Time	Activity
9.00-9.20	Welcome and introduction
9.20-10.20	Show and Tell. Each author will have three minutes to introduce their papers, provocations, or artefacts they have created for discussion (e.g. board games, cards, speculative products).
10.20-11.00	Coffee break
11.00-12.00	Keynote presentation. Stanley Greenstein, will focus on the interactions between regulatory aspects and innovation, and the multi-faceted meanings of concepts such as regulation and the law
12.00-13.00	Lunch
13.00-14.30	Mapping exercise: Drawing on key concepts outlined in the morning, we will map out different configurations of encounters between regulatory aspects, sustainability and design research, as well as the legal barriers and triggers for sustainable design
14.30-15.00	Coffee break
15.00-16.30	Future explorations: We will explore concepts, methods, or design tools that can help to more consciously work with regulatory aspects within sustainable CSCW and HCI.
16.30-17.00	Participants will socialise while discussing future actions.

Table 1: Proposed workshop schedule for the whole day

fictions. We welcome historical analysis, reports of court cases, concrete examples of regulations, theoretical pieces, or short proposals for future research. We encourage authors to outline their interest in the workshop themes. Submissions will be reviewed by the organisers and chosen based on their quality, novelty, and relevance to the workshop's topics. We aim for a good balance of perspectives and cases. We will do our best to accommodate specific *accessibility needs*. Authors will be invited to specify any particular requirement in their position papers, or by directly contacting the organisers.

4 WORKSHOP STRUCTURE

The workshop is planned as a one-day event with presentations, analytical and methodological explorations, and design-focused discussions. See Table 1 for an overview of the planned activities.

5 Post-workshop plans

The workshop aims at building a network of researchers, practitioners, and other actors interested in exploring how encounters with the law and other regulations can configure design and designing. With a focus on the relations between environmental, economic, and social sustainability, the workshop will be a platform to develop collaborations, including the organisation of future workshops or other joint activities. We plan for an article in the ACM Interactions magazine to disseminate the workshop's outcomes and point to future research opportunities. Participants' suggestions on future actions will be discussed at the workshop.

6 ORGANIZERS

The organisers' work spans sustainable interaction design, digital civics, and studies of law and IT.

Chiara Rossitto is Associate Professor of HCI at Stockholm University. Her work explores questions related to more sustainable futures, social change, and the politics of design. She is the principal investigator of a project investigating how encounters with the law and other regulations can be considered in designing for environmental sustainability.

Rob Comber is an Associate Professor at the KTH, Stockholm. He works across issues of social and environmental sustainability from a critical feminist perspective.

Anton Poikolainen Rosén is a postdoctoral researcher in HCI at Stockholm University. His research focuses on designing for sustainable futures and the more-than-human world. His most recent project focuses on regulation in processes of designing for municipal waste management.

Stanley Joel Greenstein is an Associate Professor in Law and Information Technology at the Faculty of Law, Stockholm University. In examining the relationship between information technology and law, Stanley's research and teaching have focused on the challenges associated with regulating emerging digital technologies, such as Artificial Intelligence, and on how to embed legal values already at the design.

Fatemeh Bakhshoudeh is a PhD student in HCI at the KTH in Stockholm. Her work focuses on the intersection of sustainable HCI, regulatory frameworks and, innovation within waste management.

Lachlan Urquhart is a Senior Lecturer in Technology Law and HCI at the University of Edinburgh. His work examines the regulation and design of emerging information technologies such as artificial intelligence and the Internet of Things. He leads an interdisciplinary project researching how to build more sustainable consumer electronics. He leads the Regulation and Design Lab at the Institute of Design Informatics.

Susan Lechelt is a Lecturer in Design Informatics at the University of Edinburgh. Her research explores how to design interventions, tools and prototypes to discuss the links between computing, climate change and sustainability. She is Co-I on a project exploring the Right to Repair and Internet of Things, and PI on a project exploring how sustainability concepts might be taught in the computing curriculum.

Acknowledgments

This workshop is connected to the projects "Regulating Environmental Sustainability: Encountering the Law in Designing for Waste Management", grant number 20230106, funded by the Kamprad

Family Foundation, and the "Fixing the Future" project, funded by UKRI EPSRC, grant number EP/W024780/1.

References

- [1] [n.d.]. EU EcoDesign Rules. 2023. Commission Regulation (EU) 2023/1670 of 16 June 2023, laying down ecodesign requirements pursuant to Directive 2009/125/EC of the European Parliament and of the Council and amending Commission Regulation (EU) 2023/826 (Text with EEA relevance) (2023).
- [2] Kars Alfrink, Ianus Keller, Neelke Doorn, and Gerd Kortuem. 2023. Contestable Camera Cars: A Speculative Design Exploration of Public AI That Is Open and Responsive to Dispute. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 8, 16 pages. doi:10.1145/3544548.3580984
- [3] Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmittell. 2021. On the Dangers of Stochastic Parrots: Can Language Models Be Too Big. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency* (FAccT '21). Association for Computing Machinery, New York, NY, USA, 610–623. doi:10.1145/3442188.3445922
- [4] Eli Blevis. 2007. Sustainable Interaction Design: Invention & Disposal, Renewal & Reuse. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems - CHI '07*. ACM Press, San Jose, California, USA, 503. doi:10.1145/1240624.1240705
- [5] Christina Bremer, Bran Knowles, and Adrian Friday. 2022. Have We Taken On Too Much?: A Critical Review of the Sustainable HCI Landscape. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (CHI '22). Association for Computing Machinery, New York, NY, USA, 1–11. doi:10.1145/3491102.3517609
- [6] Adrian K. Clear, Kirstie O'Neill, Adrian Friday, and Mike Hazas. 2016. Bearing an open "Pandora's box": HCI for reconciling everyday food and sustainability. *ACM Transactions on Computer-Human Interaction* 23, 5 (oct 2016), 1–25. doi:10.1145/2970817
- [7] Rob Comber and Elina Eriksson. 2023. Computing as Ecocode. In *Ninth Computing within Limits 2023*. LIMITS. <https://limits.pubpub.org/pub/a8h46wqy>.
- [8] Rob Comber and Chiara Rossitto. 2023. Regulating Responsibility: Environmental Sustainability, Law, and the Platformisation of Waste Management. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (CHI '23). Association for Computing Machinery, New York, NY, USA, 1–19. doi:10.1145/3544548.3581493
- [9] Madeleine I. G. Daep, Alex Cabral, Tiffany M Werner, Raed Mansour, Charlie Catlett, Asta Roseway, Chuck Needham, Nneka Udeagbala, and Scott Counts. 2023. The "Three-Legged Stool": Designing for Equitable City, Community, and Research Partnerships in Urban Environmental Sensing. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, Hamburg Germany, 1–19. doi:10.1145/3544548.3581289
- [10] Tawanna Dillahunt, Airi Lampinen, Jacki O'Neill, Loren Terveen, and Cory Kendrick. 2016. Does the Sharing Economy Do Any Good?. In *Proceedings of the 19th ACM Conference on Computer Supported Cooperative Work and Social Computing Companion (CSCW '16 Companion)*. Association for Computing Machinery, New York, NY, USA, 197–200. doi:10.1145/2818052.2893362
- [11] Paul Dourish. 2010. HCI and environmental sustainability: The politics of design and the design of politics. In *DIS 2010 - Proceedings of the 8th ACM Conference on Designing Interactive Systems*. ACM Press, New York, New York, USA, 1–10. doi:10.1145/1858171.1858173
- [12] Felix Anand Epp, Tim Moesgen, Antti Salovaara, Emmi Pouta, and İdil Gaziulusoy. 2022. Reinventing the Wheel: The Future Ripples Method for Activating Anticipatory Capacities in Innovation Teams. In *Designing Interactive Systems Conference* (New York, NY, USA, 2022-06-13) (DIS '22). Association for Computing Machinery, 387–399. doi:10.1145/3532106.3534570
- [13] Jennifer Gabrys, Helen Pritchard, and Benjamin Barratt. 2016. Just good enough data: Figuring data citizenships through air pollution sensing and data stories. *Big Data & Society* 3, 2 (2016), 1–14. doi:10.1177/2053951716679677
- [14] Tarleton Gillespie. 2018. *Custodians of the Internet: Platforms, Content Moderation, and the Hidden Decisions That Shape Social Media*. Yale University Press.
- [15] Elizabeth Goodman. 2009. Three Environmental Discourses in Human-Computer Interaction. In *CHI '09 Extended Abstracts on Human Factors in Computing Systems (CHI EA '09)*. Association for Computing Machinery, New York, NY, USA, 2535–2544. doi:10.1145/1520340.1520358
- [16] Stanley Greenstein. 2021. Preserving the rule of law in the era of artificial intelligence (AI). *Artificial Intelligence and Law* 30, 3 (July 2021), 291–323. doi:10.1007/s10506-021-09294-4
- [17] Lon Åke Erni Johannes Hansson, Teresa Cerratto Pargman, and Daniel Sapiens Pargman. 2021. A Decade of Sustainable HCI: Connecting SHCI to the Sustainable Development Goals. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, 1–19. doi:10.1145/3411764.3445069
- [18] Ellie Harmon and M. Six Silberman. 2018. Rating Working Conditions on Digital Labor Platforms. *Computer Supported Cooperative Work* 27, 3–6 (Dec. 2018), 1275–1324. doi:10.1007/s10606-018-9313-5
- [19] Hanna Hasselqvist, Mia Hesselgren, and Cristian Bogdan. 2016. Challenging the car norm: Opportunities for ICT to support sustainable transportation practices. In *Conference on Human Factors in Computing Systems - Proceedings*. Association for Computing Machinery, San Jose, CA, USA, 1300–1311. doi:10.1145/2858036.2858468
- [20] Steven J. Jackson, Tarleton Gillespie, and Sandy Payette. 2014. The Policy Knot: Re-Integrating Policy, Practice and Design in Cscw Studies of Social Computing. In *Proceedings of the 17th ACM Conference on Computer Supported Cooperative Work & Social Computing (CSCW '14)*. Association for Computing Machinery, New York, NY, USA, 588–602. doi:10.1145/2531602.2531674
- [21] Qing Ke. 2017. Service Providers of the Sharing Economy: Who Joins and Who Benefits? *Proceedings of the ACM on Human-Computer Interaction* 1, CSCW (Dec. 2017), 57:1–57:17. doi:10.1145/3134692
- [22] Bran Knowles, Oliver Bates, and Maria H'a akansson. 2018. This Changes Sustainable HCI. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. ACM, New York, NY, USA, 471:1–471:12. doi:10.1145/3173574.3174045
- [23] Lawrence Lessig. 2006. *Code*. Basic Books, London, England.
- [24] Max Liboiron. 2021. *Pollution Is Colonialism*. Duke University Press, Durham.
- [25] Sharon Lindberg, Chiara Rossitto, Ola Knutsson, Petter Karlström, and Sirkku Männikkö Barbuti. 2024. Doing Good Business? Design Leaders' Perspectives on Ethics in Design. *Proc. ACM Hum.-Comput. Interact.* 8, GROUP, Article 2 (Feb. 2024), 22 pages. doi:10.1145/3633067
- [26] Martin Valdemar Anker Lindrup, Arjun Rajendran Menon, and Aksel Bjørn-Hansen. 2023. Carbon Scales: Collective Sense-making of Carbon Emissions from Food Production through Physical Data Representation. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. ACM, Pittsburgh, PA, USA, 1515–1530. doi:10.1145/3563657.3596043
- [27] Orly Lobel. 2016. The Law of the Platform. In *Minnesota Law Review* 101, 88–166.
- [28] Marie Leth Meilvang. 2021. Sewage Systems, Treatment Plants, 'Blue-Green Solutions': The Role of Professionals in the Historical Justifications and Planning of Urban Wastewater Infrastructures. *European Journal of Cultural and Political Sociology* 8, 1 (Jan. 2021), 59–80. doi:10.1080/23254823.2020.1745660
- [29] Midas Nouwens, Ilaria Liccardi, Michael Veale, David Karger, and Lalana Kagal. 2020. Dark Patterns after the GDPR: Scraping Consent Pop-ups and Demonstrating Their Influence. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3313831.3376321
- [30] Roel Roscam Abbing and Ann Light. 2024. Make Friends Not Art: Mapping Law, Power and Participation in Designing an Online Platform during documenta fifteen. In *Proceedings of the Participatory Design Conference 2024: Full Papers - Volume 1* (Sibu, Malaysia) (PDC '24). Association for Computing Machinery, New York, NY, USA, 86–97. doi:10.1145/3666094.3666107
- [31] Chiara Rossitto, Rob Comber, Jakob Tholander, and Mattias Jacobsson. 2022. Towards Digital Environmental Stewardship: The Work of Caring for the Environment in Waste Management. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. Association for Computing Machinery, New York, NY, USA, 1–16. doi:10.1145/3491102.3517679
- [32] Chiara Rossitto, Martin Valdemar Anker Lindrup, Rob Comber, Jakob Tholander, Mattias Jacobsson, Alex Cabral, and Rikke Hagensby Jensen. 2023. Data-Enabled Sustainability: The Collective Work of Turning Data into Actions for Environmental Care. In *Companion Publication of the 2023 Conference on Computer Supported Cooperative Work and Social Computing (CSCW '23 Companion)*. Association for Computing Machinery, New York, NY, USA, 506–511. doi:10.1145/3584931.3611278
- [33] Robert Soden. 2022. Reimagining environmental data. *Interactions* 29, 1 (jan 2022), 45–47. doi:10.1145/3501302
- [34] Yolande Strengers. 2013. *Smart Energy Technologies in Everyday Life: Smart Utopia?* Palgrave Macmillan, New York.
- [35] Yolande Strengers, Mike Hazas, Larissa Nicholls, Jesper Kjeldskov, and Mikael B. Skov. 2020. Pursuing pleasure: Interrogating energy-intensive visions for the smart home. *International Journal of Human-Computer Studies* 136 (2020), 102379. doi:10.1016/j.ijhcs.2019.102379
- [36] Philip Tchernavskij and Susanne Bødker. 2022. Entangled Artifacts: The Meeting Between a Volunteer-run Citizen Science Project and a Biodiversity Data Platform. In *Nordic Human-Computer Interaction Conference (Aarhus, Denmark) (NordiCHI '22)*. Association for Computing Machinery, New York, NY, USA, Article 35, 13 pages. doi:10.1145/3546155.3546682
- [37] Lachlan D Urquhart, Susan Lechelt, Christopher Boniface, Haili Wu, Anna Marie Rezk, Nidhi Dubey, Melissa Terras, and Ewa Luger. 2024. The Right to Repair (R2R) Cards: Aligning Law and Design For A More Sustainable Consumer Internet of Things.. In *Nordic Conference on Human-Computer Interaction*. ACM, Uppsala Sweden, 1–20. doi:10.1145/3679318.3685341
- [38] Lachlan D Urquhart, Susan Lechelt, Melissa Terras, Neelima Sailaja, Anna Marie Rezk, Teresa Castle-Green, Dimitrios Paris Darzentas, Namrata Primalani, Violet

Owen, and Michael Stead. 2024. Creating Sustainable Internet of Things Futures: Aligning Legal and Design Research Agendas.. In *Companion Publication of the 2024 ACM Designing Interactive Systems Conference (DIS '24 Companion)*. Association for Computing Machinery, New York, NY, USA, 372–376. doi:10.

1145/3656156.3658391
[39] Ron Wakkary, Audrey Desjardins, Sabrina Hauser, and Leah Maestri. 2013. A sustainable Design Fiction: Green Practices. 20, 4 (2013), 23:1–23:34. doi:10.1145/2494265